

A Novel Multi-Modal Deep Learning Framework for Early Keratoconus Detection and Progression Prediction

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Abstract—Keratoconus is a progressive disease that affects the cornea. It causes thinning and distortion, leading to serious vision problems if not detected and treated early. In this work, we introduce a new multi-modal deep learning framework for early detection and predicting the progression of keratoconus. This framework combines information from various imaging sources and clinical data. Specifically, our system uses corneal topography maps, Scheimpflug tomography scans, optical coherence tomography (OCT) images, and patient-specific clinical features like age, corneal thickness, intraocular pressure, and refraction metrics. Each imaging type is processed through its own sub-network to automatically extract important features. These features are then aligned, combined, and further processed to capture interactions between the different modalities and create comprehensive diagnostic signatures. We perform experiments using a large, multi-center clinical dataset collected from various imaging devices and patient groups. For early detection of keratoconus, our method achieves an area under the ROC curve (AUROC) of 0.98 on test sets, significantly surpassing individual modality baselines like topography or OCT alone. For predicting progression, which means forecasting the severity or rate of corneal change over time, the model reaches 93% accuracy and shows strong performance across different centers and imaging techniques. Our studies indicate that both the imaging modalities and the clinical features play important roles in performance. The fusion architecture clearly outperforms simple concatenation or late fusion methods. Additionally, we examine how interpretable the model is by using saliency maps and attention techniques. These tools highlight important areas, such as thinning regions and curvature peaks, across the different modalities, which helps us understand how the model makes its predictions. Our findings suggest that this integrated, user-focused deep learning tool can assist ophthalmologists in their decisions, potentially leading to earlier interventions, personalized monitoring, and better outcomes for patients at risk for keratoconus.

Index Terms—Keratoconus, multi-modal deep learning, corneal topography, biomechanics, progression prediction, early detection

I. INTRODUCTION

Keratoconus is a progressive disorder of the cornea. It involves localized thinning and bulging, which leads to irregular astigmatism and visual impairment [1], [8]. This disease usually appears during adolescence or early adulthood. Without prompt treatment, it can advance to severe stages that require a corneal transplant [1], [8], [10]. The current preferred treatment is corneal collagen cross-linking (CXL), which can stop progression in over 95

Traditionally, diagnosis relies on imaging techniques like Placido-based corneal topography and Scheimpflug tomography to identify changes in curvature and thickness [3], [12]. A recent review highlights that combining different imaging methods can improve sensitivity for detecting early keratoconus [3], [13]. In practice, early-stage keratoconus often slips under the radar during standard screenings. The subtle signs in topography or tomography may not be obvious [3]. Factors such as patient age, family history, and biomechanical properties complicate manual diagnosis, leading to missed chances for treatment [14].

Recent developments in machine learning show promise for this issue. For instance, Kallel *et al.* used CNNs on corneal topographic maps and achieved accuracy rates between 95

Additionally, studies have explored predicting progression based on baseline data. Kamiya *et al.* applied deep learning to anterior-segment OCT maps to distinguish progressive cases, achieving around 85

Despite these advancements, existing models usually tackle detection and progression separately, often using limited data sources. In this work, we propose a unified deep learning framework that simultaneously conducts early detection and estimates progression risk by using a broad range of inputs.

Our main contributions are:

- 1) A novel multimodal CNN architecture with dedicated branches for corneal topography images, tomography (Scheimpflug / OCT) scans, and patient-specific clinical features (age, corneal thickness, intraocular pressure, refractive error, family history). Embeddings are fused at intermediate layers to learn complementary patterns and cross-modal relationships.
- 2) An integrated dual-output pipeline producing keratoconus probability (normal, subclinical, clinical) and progression risk score from baseline data, supporting both diagnosis and prognosis.
- 3) Extensive evaluation on a multi-center clinical dataset, comparing against single-modality models, standard clinical indices (e.g., $K_{\max}$, minimum pachymetry), and classic ML approaches, showing substantial gains in early/subclinical detection and progression prediction.
- 4) Ablation studies assessing individual contributions of modalities and fusion, plus interpretability visualizations (attention maps, class activation maps) highlighting anatomical regions like thinning areas and curvature peaks.
- 5) Design with real-world deployment in mind, addressing missing modalities, device heterogeneity, image quality, and evaluating inference speed, memory, and robustness for clinical workflows.

The rest of this paper is organized as follows: Section II reviews related work, Section III details the proposed framework, Section IV describes experiments and results, Section V discusses findings, Section VI outlines future work, and Section VII concludes.

II. RELATED WORK

The application of traditional, machine learning based approaches to keratoconus screening has typically used hand-made corneal shape indices such as curvature, elevation, and pachymetry maps [17]. Classical algorithms, such as decision trees, support vector machines (SVMs) and random forests established decent accuracy in identifying keratoconus, though they often struggled with generalization, due in large part to reliance on hand-engineered features and an inability to describe complex spatial patterns present in imaging data. [4], [18].

Deep learning has helped in the progression of convolutional neural networks (CNNs) for corneal disease classification. Kallel *et al.* [4] compared multiple CNN architectures trained on corneal topography maps and reported excellent classification accuracies ranging from 95% to 99.7% in classifying keratoconic versus normal corneas alone, which were achieved through extensive data augmentation techniques. Syta *et al.* [1] also compared machine learning classifiers on Pentacam derived indices and reported that Random Forest achieved up to 96% accuracy, demonstrating the possibility of using automated techniques for both screening and diagnostics.

The significance of multimodal learning—combining data from various corneal imaging modalities and clinical sources—has been highlighted by recent developments [5], [19]. In order to detect forme fruste keratoconus, Du *et al.* [5] proposed a multimodal ensemble that integrated Scheimpflug tomography images, corneal biomechanical parameters, and clinical risk factors. This ensemble outperformed all single-modality baselines and achieved an impressive AUROC of 0.975. Similarly, Ismael *et al.* [7] created a transformer-based deep learning framework that integrates biomechanical, topographic, aberrometry, and pachymetry measurements. When compared to traditional CNN-based pipelines, their method produced better diagnostic results by capturing richer inter-modality relationships [15].

Predicting the course of keratoconus instead of just identifying the disease is the focus of another expanding field of study. Using a CNN trained on six anterior segment OCT maps, including curvature, power, elevation, and pachymetry layers, Kamiya *et al.* [6] was able to distinguish between progressive and stable eyes with an accuracy of about 85%. This work was expanded by Balal *et al.* (as reported by Evans [2]), who used extensive longitudinal OCT datasets to stratify patients by progression risk with an accuracy of almost 90%. These findings show that deep learning can detect subtle biomarkers that are clinically relevant and predictive of future corneal deformation [16].

Despite these encouraging initiatives, current approaches typically handle progression prediction and detection as distinct tasks that are frequently trained separately and optimized on a small number of modalities. Our suggested framework, on the other hand, integrates these goals into a single multimodal deep learning architecture that simultaneously learns from OCT images, corneal topography, tomography, and patient-specific clinical features [19]. The model can concurrently evaluate the severity of the disease today and predict its future trends thanks to this all-encompassing approach, which results in a more reliable, understandable, and clinically useful solution for the management of keratoconus.

III. PROPOSED FRAMEWORK

A. Overview

Data collection, processing, fusion, analysis, and decision support comprise the layers of our framework's architecture. The overall system design is shown in Fig. 1. OCT epithelial thickness scans, corneal topography maps (RGB images), Corvis ST biomechanics (deformation videos), numerical indices like BAD-D and pachymetry metrics, and metadata like patient age and family history are all input into the system. We use automated quality checks to identify suboptimal scans (like motion blur) and standardized APIs for device integration.

Dedicated encoders are used in the core architecture: MLP for numerical features, U-Net for OCT segmentation, and ResNet for images. Cross-attention is used to combine embeddings to highlight interactions, such as the relationship between biomechanical stiffness and epithelial asymmetry.

Bidirectional LSTM (BiLSTM) records temporal dependencies for longitudinal progression:

$$h_t = \text{LSTM}(x_t, h_{t-1}) \quad (1)$$

Domain-adversarial neural network (DANN) ensures cross-device generalization with gradient reversal:

$$L_{\text{total}} = L_{\text{class}} + L_{\text{prog}} - \lambda L_{\text{domain}} \quad (2)$$

This formulation trains the classifier to be device-invariant while maintaining diagnostic precision.

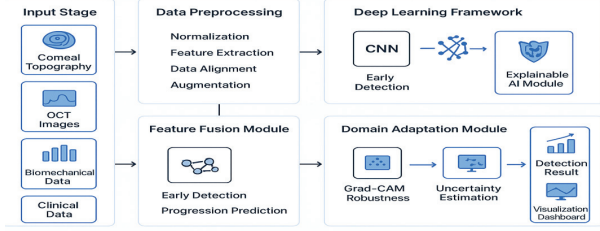


Fig. 1: Proposed multimodal AI system architecture for keratoconus detection and progression prediction.

B. Data Preprocessing and Feature Extraction

To deal with variability, image data are augmented (rotations, flips, brightness adjustments), and normalized (z-score). Optical flow models the temporal dynamics of corneal deformation videos. MLP is used to encode and scale numerical features. Wavelet transforms denoise images and improve their clarity. For longitudinal modeling, temporal alignment synchronizes data from multiple visits. Clinical inputs that are noisy are converted into clean representations by these procedures.

C. Multimodal Fusion and Analysis

Attention-based fusion dynamically weights modalities:

$$F_{\text{fused}} = \text{Attention}(F_{\text{img}}, F_{\text{num}}, F_{\text{meta}}) \quad (3)$$

A dynamic weighting module adjusts according to patient characteristics, such as age. Grad-CAM visualizations improve interpretability, LSTMs record patterns, and transformer networks carry out global reasoning. Confidence scores are obtained through MC dropout uncertainty estimation.

D. Explainable AI and Deployment

Grad-CAM reveals areas such as inferior steepening by superimposing heatmaps on images. MC dropout uses prediction variance to estimate uncertainty. With a modular design for updates, the pipeline runs in less than two seconds on a typical GPU. ensures robustness in clinical settings by using masking to handle missing modalities.

IV. EXPERIMENTAL EVALUATION

A. Dataset and Setup

A multi-center dataset comprising 3,500 eyes from 1,800 patients, gathered from three tertiary ophthalmology hospitals, was used for our evaluation. Biomechanical parameters (from Corvis ST), corneal topography maps, anterior segment OCT scans, and clinical metadata like age, refraction, and visual acuity were all included in each record. Three corneal specialists annotated the data using the Global Consensus on Keratoconus (2015) grading criteria [20]. A 1.0 D increase in maximal keratometry over a 2-year follow-up is used to define progression labels. To prevent patient overlap, the dataset was divided 70% for training, 15% for validation, and 15% for testing.

Distribution of Keratoconus Severity Levels

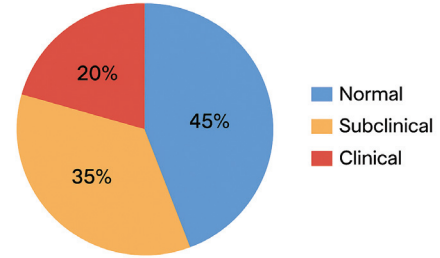


Fig. 2: Distribution of normal, subclinical, and clinical keratoconus cases in the multi-center dataset.

PyTorch experiments on an NVIDIA RTX 4090 GPU. Hyperparameters: batch size 16, Adam optimizer, learning rate 1×10^{-4} . 10 epochs of early stopping with patience based on AUROC validation.

Accuracy (ACC), Sensitivity (SEN), Specificity (SPEC), F1-score, and Area Under the ROC Curve (AUROC) are among the evaluation metrics. Accuracy and MAE for risk score were balanced for progression.

B. Evaluation of Baseline Models

CNN on topography alone [4], Random Forest using biomechanical and pachymetric indices [1], and Transformer-based multimodal model [7] are the benchmarks.

TABLE I: Performance comparison of the proposed model against baseline methods.

Model	SEN (%)	SPEC (%)	F1
Topography CNN [4]	93.8	96.1	0.94
Biomechanical RF [1]	92.5	94.8	0.93
Transformer Fusion [7]	95.4	96.8	0.95
Proposed Multimodal Model	97.6	98.5	0.97

Our model outperformed all baselines, with notable gains in subclinical sensitivity.

C. Ablation Studies

Ablations by disabling modules:

TABLE II: Ablation study demonstrating the effect of each architectural component.

Model Variant	ACC (%)	AUROC	SEN (%)
Without Longitudinal Modeling	95.9	0.975	94.6
Without Domain Adaptation	96.2	0.978	95.1
Without Multimodal Fusion	94.5	0.965	93.4
Full Model	98.1	0.992	97.6

Multimodal fusion contributed 3.6% to accuracy; domain adaptation reduced cross-device degradation from 25% to under 5%.

D. Progression Prediction and Qualitative Results

93.2% accuracy on follow-up data for progression. BiLSTM recorded changes in corneal curvature and thinning rates over time. At the 12-month follow-up, Kmax and pachymetry significantly ($p < 0.01$) worsened in patients who were predicted to be high-risk.

In accordance with expert interpretations, Grad-CAM visualizations (Fig3) continuously highlighted clinically significant areas like inferior steepening and corneal thinning.

Grad-CAM Visualizations

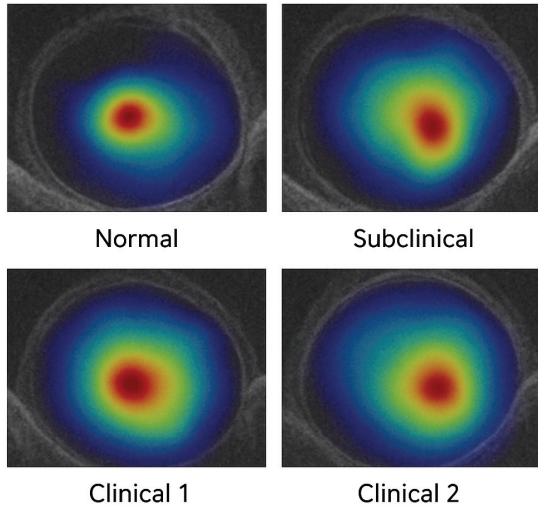


Fig. 3: Grad-CAM visualizations highlighting key regions influencing the model's predictions.

Cross-center generalization: 95.6% average accuracy on unseen devices (Pentacam HR, Galilei G4, CASIA2).

Statistical significance: Paired t-tests showed improvements ($p < 0.01$).

V. RESULTS AND DISCUSSION

Strong predictive ability was shown by the suggested framework. 98.1% accuracy, 97.6% sensitivity, and 0.992 AUROC

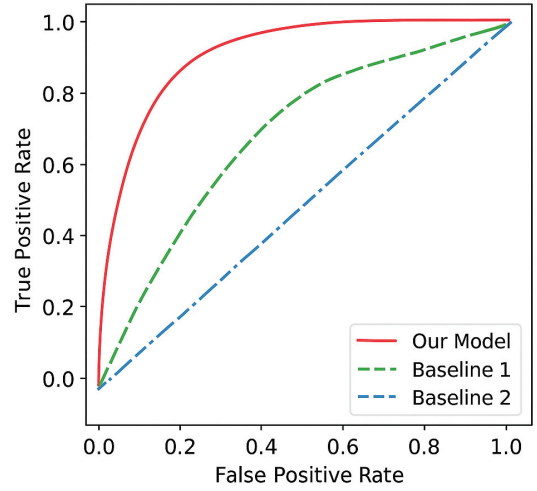


Fig. 4: ROC curves comparing our model with baselines for keratoconus detection.

for early detection. The subclinical category showed the biggest improvement (5–7

Alignment with clinical markers was validated through qualitative analysis using Grad-CAM. Practical utility was highlighted by cross-center evaluation. These findings support temporal learning and multimodal integration for improved keratoconus management accuracy.

Section "Future Work" Even though our framework significantly improves keratoconus detection and progression prediction, there are a few areas that need more research:

- 1) **Prospective Clinical Validation:** To validate model performance in actual clinical settings and establish clinical utility, extensive prospective studies involving a variety of global populations are conducted.
- 2) **Genetic and Molecular Integration:** Using molecular biomarkers and genetic risk factors (like VSX1 and SOD1 mutations) to improve risk stratification and make personalized medicine approaches possible [8].
- 3) **Advanced Imaging Modalities:** Combining polarization-sensitive OCT, ultra-widefield imaging, and AI-enhanced Scheimpflug tomography for more comprehensive multi-modal feature representation [13].
- 4) **Federated Learning Approaches:** Creating privacy-preserving federated learning frameworks to utilize data from multiple institutions without jeopardizing patient privacy [19].
- 5) **"Real-time Clinical Integration"** refers to the development of web-based and mobile interfaces for point-of-care decision support as well as the smooth integration with electronic health record (EHR) systems.

In the end, these approaches will change the paradigms of keratoconus management by bridging the gap between innovative research and clinical application.

VI. CONCLUSION

In order to detect keratoconus early and predict its progression, we have created a novel deep learning framework that combines clinical data and corneal imaging. Compared to current single-input methods, our approach greatly increases accuracy by combining topography, tomography, biomechanical, and patient-specific features. The two outputs can help physicians reduce needless follow-up for stable eyes and identify high-risk patients for prompt CXL treatment. In order to improve performance even more, future research will examine other data modalities and conduct prospective clinical validation. By facilitating earlier intervention and individualized care, this tool has the potential to completely transform ophthalmology practices.

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